

Architecture, Agents, Automation, and Operating Notes

Prepared for the Khoj STEAM School execution stack. Version 1.0, 31 May 2026.

Purpose

KhojAI is the automated work engine behind Khoj STEAM School. It is intended to generate educational content, run support agents, maintain a WhatsApp teacher experience, store conversations and lesson state, and publish outputs for the public Khoj product experience.

Public product experience: <https://khoj.zuhaib.pro/> | Internal tracking dashboard: <https://lab.zuhaib.pro/khoj>

Known Deployment Shape

Area	Status	Notes
/var/www/khojai	VPS expected	Server-side project root provided as the live KhojAI work engine.
Local mirror	Confirmed locally	khoj/claude - code/khojai/khojai contains scheduler, package metadata, Baileys teacher, and old Meta bot.
Website process	VPS expected	Next.js website is configured for port 3001 through next start -p 3001.
Teacher bot	Confirmed locally	Baileys WhatsApp teacher reads and writes khoj.db, uses Claude, and handles text plus voice transcription.

Folder Map

/var/www/khojai/	
-- app/	Next.js website
-- agents/	AI agents: curriculum, content, visual, SEO, QA, director
-- aiteacher/	WhatsApp AI teacher using Baileys
-- bot/	Old Meta Cloud API bot, currently legacy
-- content/output/	Generated images, carousel copy, lesson payloads
-- logs/	Scheduler, pipeline, bot, and agent logs
-- pipeline.js	Older pipeline entrypoint
-- scheduler.js	Current scheduler, every 2 days plus support runs
-- khoj.db	SQLite database for content, users, and conversations
-- .env	API keys and runtime configuration
-- next.config.mjs	Next.js configuration
-- package.json	Node dependencies and scripts

Local inspection confirms scheduler.js, package.json, aiteacher/index.js, and bot/teacher.js. The local mirror does not currently expose the full app/, agents/, content/output/, pipeline.js, or database files.

Runtime Components

Component	What It Does
Next.js Website	Serves the public KhojAI web experience. Scripts define dev, build, and start on port 3001.

Component	What It Does
Scheduler	Uses node-cron. Runs content pipeline every two days at midnight and support agents Monday and Thursday at 3 AM.
AI Teacher	Uses WhatsApp Baileys session auth, Claude for replies, OpenAI Whisper for voice transcription, and SQLite for users/history.
Legacy Meta Bot	Express webhook implementation for Meta Cloud API. Useful as reference; active teacher path appears to be aiteacher/index.js.

Scheduler Behavior

Trigger	Schedule	Action
Content pipeline	0 0 */2 * *	Runs runContentPipeline() every two days at midnight.
Support agents	0 3 * * 1,4	Runs runSupportAgents() every Monday and Thursday at 3 AM.
Manual pipeline	node scheduler.js --now	Immediately runs the content pipeline.
Manual support	node scheduler.js --support	Immediately runs support agents.

Database Notes

Table	Purpose
wa_conversations	Stores WhatsApp user and assistant message history.
wa_users	Stores phone, name, relation, child profile, language, and message count.
conversations	Used by the legacy Meta bot for webhook conversation history.
students	Used by the legacy Meta bot for parent/student profile data.
content	Referenced by both teacher paths for latest lesson context. Schema should be verified on VPS.

Dependency Snapshot

Dependency	Role
next, react	Public website UI.
node-cron	Scheduled automation.
@anthropic-ai/sdk	Claude-powered teacher and agents.
@google/generative-ai	Gemini provider support for agent work.
@whiskeysockets/baileys	WhatsApp web session teacher.

Dependency	Role
better-sqlite3	Local SQLite storage.
axios, form-data	HTTP calls and voice transcription upload.
express	Legacy Meta bot webhook server.

Operational Risks

Risk	Why It Matters	Recommended Fix
Auth/session files in local mirror	Baileys auth files can expose session state.	Keep aiteacher/auth/ out of Git and rotate sessions if exposed.
Hardcoded absolute VPS paths	Local and VPS behavior can diverge.	Move root paths into .env with sane defaults.
Legacy and new bot both present	Can confuse deployment and duplicate behavior.	Mark bot/ as legacy or archive after verifying no process uses it.
Missing local full website and agents	Harder to review or rebuild from local source.	Sync VPS source into repo, excluding secrets, logs, database, and generated outputs.

Recommended Next Actions

Priority	Action	Outcome
P0	Create a clean Git repo for khojai with secrets excluded.	VPS and local code stop drifting.
P0	Add .gitignore for .env, khoj.db, logs/, .next/, and aiteacher/auth/.	Prevents secret/session leakage.
P1	Document PM2 process names, ports, and Nginx routes on VPS.	Clear deploy and recovery map.
P1	Verify agents/director, pipeline outputs, and database content schema on VPS.	Completes agent workflow documentation.
P2	Expose selected public outputs from KhojAI into https://khoj.zuhaib.pro/ .	Parents see product content, not only infrastructure status.

This wiki intentionally excludes API keys, tokens, database contents, WhatsApp session files, and private phone numbers.

Confirmed local source: khoj/claude - code/khojai/khojai. VPS structure based on the supplied /var/www/khojai map.